Hogwarts Division Review

Grade 5 Chapter 3 Review • Division • Page 1 of 4

Wizard's Name: ..... Date: .....

House Points ...../10

1 At Gringotts, a goblin stacks **54 Galleons** into **6** equal piles: **54 ÷ 6 = 9**. **Use the pattern to find each quotient.**

a. **540 ÷ 6** = ☐

b. **5,400 ÷ 6** = ☐

c. **54,000 ÷ 6** = ☐

d. **4,200 ÷ 70** = ☐

2 Griphook checks which numbers each vault number can be divided by evenly. **Add up the digits first. Then write Y (yes) or N (no) for each rule.**

Vault number	Digit sum	÷ 2	÷ 3	÷ 4	÷ 5	÷ 6	÷ 9	÷ 10
687								
1,260								
4,716								

3 The trolley on the Hogwarts Express sells **2,738** Chocolate Frogs on **7** trips, the same number each trip.

a. **Use compatible numbers** to estimate how many frogs each trip:

☐ ÷ ☐ = ☐

b. Ron says the exact answer is **39 R1** frogs. Use your estimate to explain why Ron is wrong.

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4 Order of operations! Professor McGonagall writes two puzzles on the board.
Simplify each one. Show every step.
Remember: ( ) first, then × ÷ left to right, then + − left to right.

a. **48 ÷ (2 + 6) × 3 − 4**

Answer

b. **36 − 18 ÷ 3 × 2 + (7 − 2)**
Draco worked left to right and got **17**. Find the right answer.

Answer

5 Hagrid plants **4,839** pumpkin seeds in **6** equal rows, then **3,248** seeds in **8** rows.
Watch for zeros! **Find each quotient and remainder, then check by multiplying.**

a. $6\overline{)4{,}839}$

Answer

Check: ☐ × ☐ + ☐ = ☐

b. $8\overline{)3{,}248}$

Answer

Check: ☐ × ☐ + ☐ = ☐

6 Uncle Vernon spends **$183.15** on Dudley's **37** birthday presents. Each present costs the same. **How much does each present cost?**
Remember the $ and the decimal point!

Show your work

$37\overline{)\$183.15}$

Answer

Remainder Riddles. Every one of these has a remainder. Read each story carefully and decide what the remainder means for the answer.
Check: your remainder must always be smaller than the divisor.

7 Across the Lake

Hagrid leads **103** first-years down to the lake. Each boat holds **4** students. **How many boats do the first-years need?**

Quotient ☐ R ☐

Answer ☐ boats

Show your work

$4 \overline{)103}$

I rounded ☐ up ☐ down because .....

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.....

8 Potions with Snape

One batch of Cure for Boils uses **12** snake fangs. Neville has **315** fangs. **How many batches can he brew?**

Quotient ☐ R ☐

Answer ☐ batches

Show your work

$12 \overline{)315}$

I rounded ☐ up ☐ down because .....

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9 Every Flavor Beans

The trolley witch has **1,150** jelly beans. She packs **24** beans into each box.

a. How many boxes can she fill?

b. How many beans are left over?

Quotient ☐ R ☐

a. Answer ☐ boxes

b. Left over ☐ beans

Show your work

24 ⟌ 1,150

I rounded ☐ up ☐ down because .....

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10 The Great Hall Candles

Filch needs **20,000** floating candles for the Welcome Feast. The shop in Hogsmeade sells candles in boxes of **36**.

How many boxes should he buy?

Quotient ☐ R ☐

Answer ☐ boxes

Show your work

36 ⟌ 20,000

I rounded ☐ up ☐ down because .....

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Answer Key

For grown-ups only: not part of the student pages

1. Patterns: **90**, **900**, **9,000**, then 4,200 ÷ 70 = **60** Lessons 3-1 and 3-2, pp. 96–99

   Use the basic fact, then count zeros. In part d, cross off one zero from each number (420 ÷ 7 = 60). Watch for 600, which comes from forgetting the zero in 70.

2. Divisibility: Lessons 3-7 and 3-8, pp. 108–111

   687: digit sum **21**; **only ÷ 3 is Y** *(21 is not divisible by 9; odd, so no 2, 4, 6)*

   1,260: digit sum **9**; **all Y** *(ends in 0; 60 ÷ 4 = 15; even and ÷ 3, so ÷ 6)*

   4,716: digit sum **18**; **Y for 2, 3, 4, 6, 9**; **N for 5, 10** *(16 ÷ 4 = 4; 18 is divisible by 9; ends in 6)*

3. Compatible numbers: Lesson 3-9, p. 112

   a. **2,800 ÷ 7 = 400** frogs *(28 is a basic fact with 7)*

   b. **39 is far too small.** The estimate is about 400, so the answer needs three digits. *(The exact answer is 391 R1. Ron's quotient is missing a digit.)*

4. Order of operations: Lesson 3-14, p. 122

   a. 48 ÷ 8 × 3 − 4 → 6 × 3 − 4 → 18 − 4 → **14**

   b. 36 − 18 ÷ 3 × 2 + 5 → 36 − 6 × 2 + 5 → 36 − 12 + 5 → 24 + 5 → **29**

   Draco subtracted 36 − 18 first. Parentheses come first, then × and ÷ from left to right, and only then + and −.

   *Also watch for **38** (did 3 × 2 before 18 ÷ 3, reading PEMDAS as "multiply before divide") and **19** (did 12 + 5 before 36 − 12, reading it as "add before subtract").*

5. Zeros in the quotient: **4,839 ÷ 6 = 806 R3** **3,248 ÷ 8 = 406** Lessons 3-5 and 3-6, pp. 104–107

   Check: 806 × 6 + 3 = 4,839 406 × 8 = 3,248

   Watch for 86 R3 and 46: the tens step has a 0 that must be written in the quotient.

6. **$183.15 ÷ 37 = $4.95** per present Lesson 3-13, p. 120

   Place the $ and the decimal point in the quotient first, then divide as usual. Check: 37 × $4.95 = $183.15.

7. **103 ÷ 4 = 25 R3** → **26 boats** (round up) Lesson 3-3, p. 100

   25 boats leave 3 students on the shore, so they need one more boat. If your child answers 27 because Hagrid rides in a boat of his own (as in the book), that is good reading: accept it if the 26 is in their work.

8. **315 ÷ 12 = 26 R3** → **26 batches** (round down) Lesson 3-10, p. 114

   The 3 leftover fangs are not enough for another batch of 12.

9. **1,150 ÷ 24 = 47 R22** Lesson 3-11, p. 116

   a. **47 boxes** (round down) b. **22 beans** left over

   The 22 leftover beans can't fill a box of 24, so they don't add a box. In part b the remainder itself is the answer.

10. **20,000 ÷ 36 = 555 R20** → **556 boxes** (round up) Lesson 3-12, p. 118

    555 boxes is only 19,980 candles, 20 short. He can't buy part of a box, so he needs one more whole box.